

Verifikasi Kalibrasi Gauge dan Transmitter



27 Agustus 2021

E&I Maintenance

PT. Multimas Nabati Asahan - Serang

Contents

01

Apa itu Gauge dan Transmitter

02

Apa itu Verifikasi?

03

Apa itu Kalibrasi?

04

Mengapa perlu dilakukan Verifikasi dan Kalibrasi?

Gauge dan Transmitter

Gauge dan Transmitter

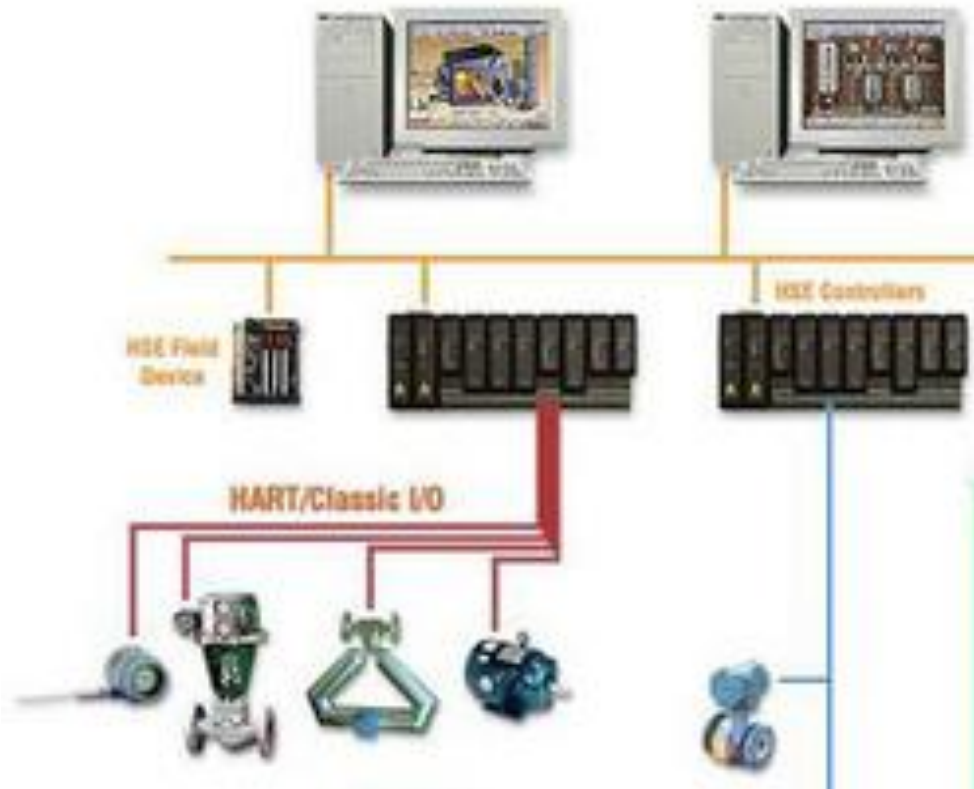
Gauge ialah sebuah peralatan instrument yang menampilkan secara langsung nilai suatu pengukuran didalam proses line

Sedangkan transmitter adalah peralatan instrument yang mengubah perubahan sensing element dari sebuah sensor menjadi sinyal yang mampu diterjemahkan oleh controller (PLC dan DCS)



Gauge dan Transmitter

Hal yang mendasar apabila gauge hanya dapat menampilkan hasil pengukuran sensor secara langsung (di area tersebut) sedangkan transmitter sendiri selain nilai hasil pengukuran dari sensor juga dapat ditampilkan ke HMI kita sebagai interface



Jenis-Jenis Pengukuran Gauge

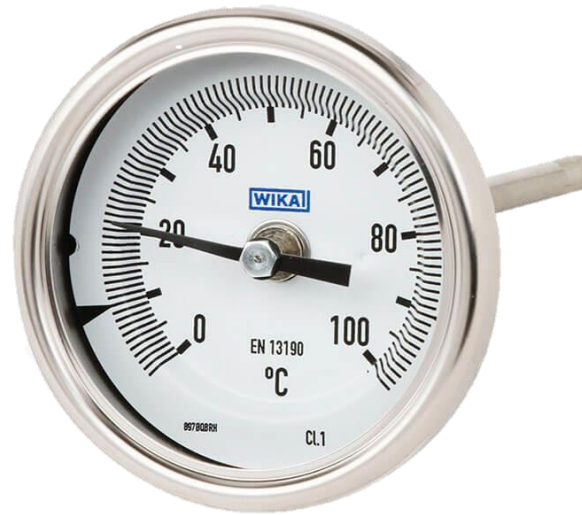
PG

Pressure
Gauge



TG

Temperature
Gauge

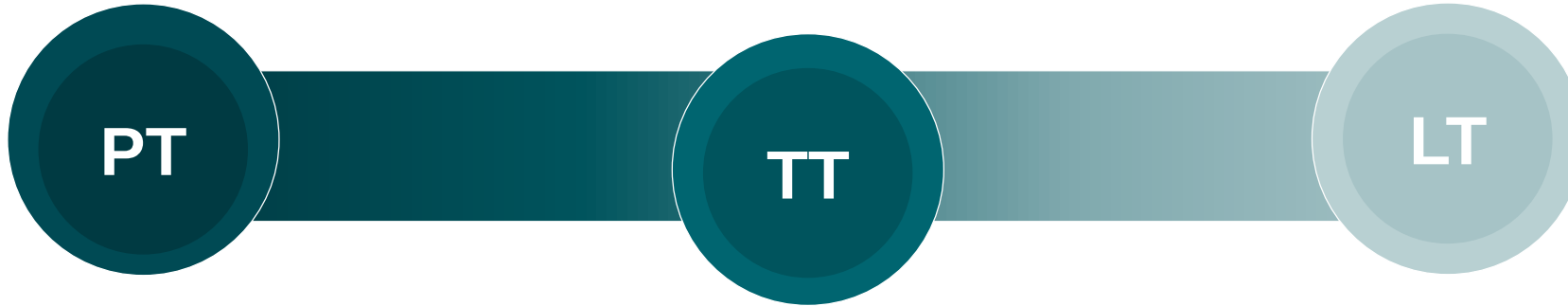


LG

Level
Gauge



Jenis-Jenis Pengukuran Transmitter



**Pressure
Transmitter**



**Temperature
Transmitter**



**Level
Transmitter**



Verifikasi dan Kalibrasi

Verifikasi



Verifikasi adalah proses “Konfirmasi” bahwa syarat dalam spesifikasi telah terpenuhi. Spesifikasi bisa didapat dari alat ataupun aturan standar tertentu

Hal yang perlu diperhatikan dalam verifikasi:

- Mengkonfirmasi performa dari suatu alat berdasarkan spesifikasi
- Memastikan peralatan tersebut bekerja sesuai fungsinya
- Tidak memeriksa seluruh range, hanya memastikan dapat berkerja di range tersebut



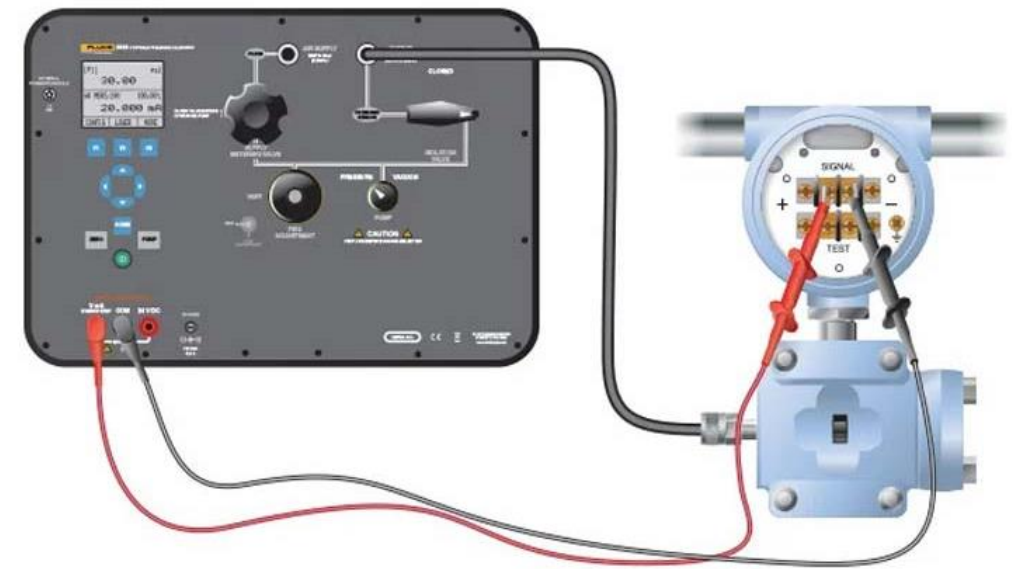
Kalibrasi

Konsep dasar Kalibrasi adalah menentukan hubungan antara nilai yang ditunjukkan oleh alat ukur atau sistem pengukuran atau bahan ukur atau bahan acuan (dalam kondisi tertentu) dengan nilai yang direalisasikan oleh standar

Nilai yang direalisasikan oleh standar = (taksiran) nilai benar

Kalibrasi itu sendiri ialah menentukan perbedaan (deviasi) antara pembacaan alat ukur atau bahan ukur (yang digunakan sebagai standar) dengan (taksiran) nilai benar

”



Secara garis besar apabila kita melakukan verifikasi kita hanya melihat berapa selisih alat yang di verifikasi dengan alat yang digunakan verifikasi, tanpa ada perhitungan eror, factor K, penyimpangan alat

Namun apabila kalibrasi kita melakukan verifikasi alat yang diverifikasi terhadap alat yang kita jadikan master dan melakukan perhitungan eror, faktor K, penyimpangan dan melakukan adjustmen apabila diperlukan atau dibutuhkan



Pentingnya Verifikasi dan Kalibrasi

Tujuan Perlu dilakukan Verifikasi dan Kalibrasi



**Apakah semua Instrument
dapat dikalibrasi ataupun
verifikasi?**



**Tidak semua instrument dapat
dilakukan verifikasi ataupun
kalibrasi seperti actuated valve,
level switch dan breather valve**

Calirbation Training

Kalibrator Fluke 9140



Spesifikasi

- 35° C to 350° C
- 230 V AC ($\pm 10\%$), 2.2 A
- 500 W
- 152 x 86 x 197 mm
- 2.7 kg

Sertifikat Kalibrasi Fluke 9140



Certificate of Calibration Fluke Park Laboratory

Description: Drywell Calibrator	Certificate Number: C0A06029
Manufacturer: Fluke	Date of Calibration: 03 Oct 2020
Model: 9140	Date Due:
Serial Number: C0A061	Temperature: 20.0 to 26.0 °C
Status: As-Found: New	Relative Humidity: 15 to 70 %RH
As-Left: In Tolerance	Pressure: 95 to 103 kPa
Calibration: Full	Issue Date: 06 Oct 2020
Procedure: AFC480 - 001	

Customer: FLUKE SOUTH EAST ASIA PTE LTD SINGAPORE, SG	RMA/SO Number: 9665645
PO Number: 19539/SG//STOCK	

This calibration is traceable to the SI through recognized national measurement institutes (NIST, PTB, NPL, NIM, NRC, etc.), radiometric techniques, or natural physical constants. The calibration has been completed in accordance with the Fluke Corporate Quality System document QSD 111.0. Calibration certificates without identification of the authorizing person are not valid. This certificate applies to only the item identified and shall not be reproduced other than in full, without the specific written approval by Fluke Corporation.

Comments:



Cert: C0A06029
Date: 03/10/2020
S/N: C0A061


Approved Signatory
Tai Nguyen

Fluke Corporation
6920 Seaway Blvd
Everett, WA 98203 USA

Telephone
877.355.3225

Internet
www.flukecal.com

Page 1 of 3
Rev 20160922

Certificate of Calibration

Model: 9140
Serial No: C0A061
Report No: C0A06029

As Found Data

No As Found Data Required

As Left Data

Calibration Constants		Test Data				
		Set-point °C	Actual °C	Error °C	Calibration Tolerance °C	Pass/Fail
R0	101.096	50.0	50.00	0.00	±0.50	P
Alpha	0.0038194	100.0	99.98	-0.02	±0.50	P
Delta	1.2583	150.0	149.96	-0.04	±0.50	P
Proportional Band	15.0	200.0	199.97	-0.03	±0.50	P
		250.0	250.01	0.01	±0.50	P
		300.0	300.07	0.07	±0.50	P
		350.0	349.98	-0.02	±0.50	P

Kalibrator Fluke 9142



Spesifikasi

- -25° C to 150° C
- 230 V AC ($\pm 10\%$)
- 630 W
- 290 x 185 x 295 mm
- 8.16 kg

Sertifikat Kalibrasi Fluke 9142



Certificate of Calibration Fluke Park Laboratory

Description:	Field Metrology Well with Process Readout	Certificate Number:	C0303011
Manufacturer:	Fluke	Date of Calibration:	03 Mar 2020
Model:	9142	Date Due:	
Serial Number:	C03087	Temperature:	20.0 to 26.0 °C
Status:	As-Found: New	Relative Humidity:	15 to 70 %RH
	As-Left: In Tolerance	Pressure:	95 to 103 kPa
Calibration:	Full	Issue Date:	03 Mar 2020
Procedure:	HCT301 - 1		
Customer:	FLUKE SOUTH EAST ASIA PTE LTD SINGAPORE, SG	RMA/SO Number:	9636976
PO Number:	18785/SG//STOCK		

This calibration is traceable to the SI through recognized national metrological institutes (NIST, PTB, NPL, NIM, NRC, etc.), radiometric techniques, or natural physical constants and is in compliance with ISO/IEC 17025:2005 and ANSI/NCSL Z540.1. The calibration has been completed in accordance with the Fluke Corporate Quality System document QSD 111.0. Calibration certificates without identification of the authorizing person are not valid. This certificate applies to only the item identified and shall not be reproduced other than in full, without the specific written approval by Fluke Corporation.

This calibration certificate may contain data that is not covered by the Scope of Accreditation. The unaccredited test points, where applicable, are indicated by an asterisk (*), or confined to clearly marked sections. This certificate shall not be used to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the U.S. Government.

Measurement uncertainties at the time of test are given where applicable. They are calculated in accordance with the method described in the ISO Guide to the Expression of Uncertainty in Measurement. The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k such that the coverage probability corresponds to approximately 95 %.

Comments:



Cert # : C0303011
Date Cal: 03 Mar 2020
Date Due:
S/N : C03087
877-201-1111 www.flukecal.com



Cert: C0303011
S/N: C03087

Approved Signatory

Amacy Rico

Certificate of Calibration

Model: 9142
Serial No.: C03087
Certificate No: C0303011

As Found Data

No As Found Data Required

As Left Data

Data ID: C0062082457031

Calibration Constants		Temperature Accuracy						
		Set-point °C	Actual °C	Error °C	Tolerance °C	Uncertainty	Pass/Fail	
TEMP 1	0.143	-25.000	-24.979	0.021	±0.200	±0.025	P	
TEMP 2	0.013	0.000	-0.019	-0.019	±0.200	±0.025	P	
TEMP 3	-0.158	50.000	49.987	-0.013	±0.200	±0.025	P	
GRAD 1	-0.014	100.000	100.032	0.032	±0.200	±0.030	P	
GRAD 2	0.017	150.000	149.992	-0.008	±0.200	±0.030	P	
GRAD 3	0.013							
		Temperature Stability						
		Set-point °C	Observed °C (2 Sigma)	Tolerance °C	Uncertainty	Pass/Fail		
Control Constants		-25.000	0.002	±0.010	±0.0040	P		
TEMP PB	2.0	150.000	0.002	±0.010	±0.0055	P		
TEMP INT	30.0							
TEMP DER	1.0							
		Axial Differential Temperature						
		Set-point °C	Target °C	Actual °C	Error °C	Tolerance °C	Uncertainty	Pass/Fail
		-25.000	0.000	0.003	0.003	±0.040	±0.020	P
		50.000	0.000	-0.008	-0.008	±0.040	±0.020	P
		100.000	0.010	0.023	0.013	±0.040	±0.025	P
		150.000	0.010	0.010	0.000	±0.040	±0.030	P

Kalibrasi Temperature

1. Lepas RTD dan Transmitter dari thermoweld

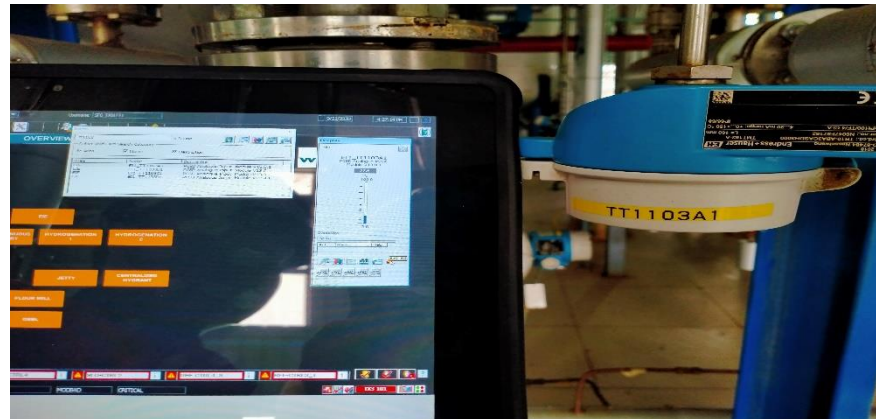


2. Pasang RTD ke fluke calibrator dan connect transmitter ke AMS Trex
3. Catat nilai suhu yang terbaca di fluke calibrator, AMS Trex dan nilai resistansi dari sebuah RTD



Kalibrasi Temperature

4. Pasang kembali RTD dan pastikan terminasi sudah kencang dan tebaca di DCS



5. Lakukan analisis data kalibrasi dan nilai penyimpangannya

Perhitungan ketidakpastian													
Penunjukan Thermometer yang dikalibrasi (°C)	Standar deviasi (S)	Tipe A		Tipe B							Drajat kebebasan Efektif (Veff)	Faktor cakupan (k)	Ketidakpastian dipertuas (U95)
		Ketidakpastian (U _A)	Drajat kebebasan (V _A)	Ketidakpastian Kalibrator (U _S)	Drajat kebebasan (V _S)	Ketidakpastian Daya Baca kalibrator (U _{DS})	Drajat kebebasan (V _{DS})	Ketidakpastian Daya Baca alat yang dikalibrasi (U _{DA})	Drajat kebebasan (V _{DA})	Ketidakpastian standar gabungan (U _C)			
25,03	0,039	0,023	2	0,306	50	0,006	50	0,006	8	0,307	51	2,03	0,6
30,02	0,036	0,021	2	0,306	50	0,006	50	0,006	8	0,307	51	2,03	0,6
35,02	0,038	0,022	2	0,306	50	0,006	50	0,006	8	0,307	51	2,03	0,6
40,02	0,030	0,018	2	0,306	50	0,006	50	0,006	8	0,307	50	2,03	0,6
45,00	0,010	0,006	2	0,306	50	0,006	50	0,006	8	0,306	50	2,03	0,6
Nilai toleransi : 1,5 °C													
Kesimpulan : Dapat digunakan karena nilai toleransinya masih lebih besar dari nilai penyimpangan.													

Kalibrator Fluke 3130



Spesifikasi

- Battery Internal 3800 mAh
- Power 12 Vdc (Charger)
- - 0.8 to 20 bar (pump)
- 7 Kg
- 1/8 NPT
- bar, mbar, psi, kPa, MPa, mmH₂O

Sertifikat Kalibrasi Fluke 3130



Certificate of Calibration Fluke Park Laboratory

Description:	PORTABLE PRESSURE CALIBRATOR	Certificate Number:	1500273517
Manufacturer:	FLUKE	Date of Calibration:	19 Nov 2019
Model:	3130	Date Due:	
Serial Number:	4780008	Temperature:	20.0 to 26.0 °C
Status:	AS-LEFT	Relative Humidity:	10 to 70 %RH
		Pressure:	95 to 103 kPa
Calibration:	FULL	Issue Date:	19 Nov 2019
Procedure:	FLUKE 3130 (5520/8508/RPM4) : 20190228		
Customer:	FLUKE SOUTH EAST ASIA PTE LTD SINGAPORE, SG	RMA/SO Number:	9620814
PO Number:	2670712-FCO-Q/ID/ /PT		

This calibration is traceable to the SI through recognized national measurement institutes (NIST, PTB, NPL, NIM, NRC, etc.), radiometric techniques, or natural physical constants and is in compliance with ISO/IEC 17025:2005 and ANSI/NCSL Z540.1. The calibration has been completed in accordance with the Fluke Corporate Quality System document QSD 111.0. Calibration certificates without identification of the authorizing person are not valid. This certificate applies to only the item identified and shall not be reproduced other than in full, without the specific written approval by Fluke Corporation.

This calibration certificate may contain data that is not covered by the Scope of Accreditation. The unaccredited test points, where applicable, are indicated by an asterisk (*), or confined to clearly marked sections. This certificate shall not be used to claim product certification, approval, or endorsement by NVLAP, NIST, or any agency of the U.S. Government.

Measurement uncertainties at the time of test are given where applicable. They are calculated in accordance with the method described in the ISO Guide to the Expression of Uncertainty in Measurement. The reported expanded uncertainty of measurement is stated as the standard uncertainty of measurement multiplied by the coverage factor k such that the coverage probability corresponds to approximately 95 %.

Comments:



Cert # : 1500273517
Date Cal: 19 Nov 2019
Date Due:
S/N : 4780008
877-500-1212 www.flukecal.com

Cert: 1500273517
Date: 19 Nov 2019
S/N: 4780008

Approved Signatory
Josh Hanger

Certificate Number: 1500273517

Date of Calibration: 19 Nov 2019

Applied	Test Result	Error	Measurement Uncertainty	Specification
Instrument Identification: FLUKE 3130 4780008 1.08				
DC VOLTAGE MEASURE				
0.0000 V	-0.001 V	-0.0010 V	5.8E-004 V	0.002 V
5.0000 V	4.999 V	-0.0007 V	8.8E-004 V	0.003 V
10.0000 V	10.000 V	0.0000 V	5.9E-004 V	0.004 V
15.0000 V	15.000 V	0.0000 V	6.0E-004 V	0.004 V
20.0000 V	20.000 V	0.0000 V	6.1E-004 V	0.005 V
25.0000 V	25.000 V	0.0000 V	6.3E-004 V	0.006 V
30.0000 V	30.000 V	0.0000 V	6.5E-004 V	0.007 V
DC CURRENT MEASURE				
4.0000 mA	4.000 mA	0.0000 mA	7.7E-007 A	0.003 mA
8.0000 mA	8.000 mA	0.0003 mA	1.2E-006 A	0.003 mA
12.0000 mA	12.000 mA	0.0000 mA	1.3E-006 A	0.004 mA
16.0000 mA	16.001 mA	0.0010 mA	1.5E-006 A	0.004 mA
20.0000 mA	20.001 mA	0.0010 mA	1.8E-006 A	0.005 mA
24.0000 mA	24.000 mA	0.0000 mA	2.1E-006 A	0.006 mA
DC CURRENT SOURCE				
0.000 mA	0.0001 mA	0.0001 mA	5.9E-007 A	0.0020 mA
4.000 mA	3.9995 mA	-0.0005 mA	6.1E-007 A	0.0026 mA
8.000 mA	8.0001 mA	0.0001 mA	6.2E-007 A	0.0032 mA
12.000 mA	12.0000 mA	0.0000 mA	6.5E-007 A	0.0038 mA
16.000 mA	15.9994 mA	-0.0006 mA	6.6E-007 A	0.0044 mA
20.000 mA	19.9998 mA	-0.0002 mA	1.4E-006 A	0.0050 mA
24.000 mA	23.9995 mA	-0.0005 mA	1.6E-006 A	0.0056 mA
LOOP VOLTAGE Loop Voltage between 23V and 28V - Measured = 25.0744				
mA SIM-2N				
4.000 mA	3.9994 mA	-0.0006 mA	6.2E-007 A	0.0026 mA
16.000 mA	15.9994 mA	-0.0006 mA	6.7E-007 A	0.0044 mA
24.000 mA	23.9996 mA	-0.0004 mA	1.6E-006 A	0.0056 mA
PRESSURE in PSI and converted to kPa Conversion Factor of 6.894759				
0.001 psi	0.00 kPa	-0.001 psi	1.6E-002 psi	0.03 psi
0.01000 kPa	0.0000 kPa	-0.01000 kPa	1.1E+002 Pa	0.2068 kPa
50.020 psi	50.01 psi	-0.010 psi	1.6E-002 psi	0.04 psi
344.88000 kPa	344.8800 kPa	-0.0000 kPa	1.1E+002 Pa	0.1930 kPa
150.013 psi	150.02 psi	0.004 psi	1.8E-002 psi	0.07 psi
1034.30000 kPa	1034.4000 kPa	0.10000 kPa	1.1E+002 Pa	0.4654 kPa
250.001 psi	250.01 psi	0.012 psi	2.7E-002 psi	0.09 psi
1723.70000 kPa	1723.8000 kPa	0.10000 kPa	1.7E+002 Pa	0.6377 kPa
274.978 psi	275.00 psi	0.022 psi	2.8E-002 psi	0.10 psi
1895.91000 kPa	1896.1000 kPa	0.19000 kPa	1.9E+002 Pa	0.6800 kPa
299.972 psi	300.00 psi	0.028 psi	3.1E-002 psi	0.10 psi
2068.23000 kPa	2068.4000 kPa	0.17000 kPa	2.1E+002 Pa	0.7239 kPa
199.998 psi	200.01 psi	0.012 psi	2.1E-002 psi	0.08 psi
1378.94000 kPa	1379.0000 kPa	0.06000 kPa	1.4E+002 Pa	0.5515 kPa
125.027 psi	125.03 psi	0.003 psi	1.6E-002 psi	0.06 psi
862.03000 kPa	862.1000 kPa	0.07000 kPa	1.1E+002 Pa	0.4223 kPa
100.029 psi	100.03 psi	-0.002 psi	1.8E-002 psi	0.06 psi
689.60000 kPa	689.6000 kPa	-0.00000 kPa	1.1E+002 Pa	0.3792 kPa
0.000 psi	0.00 psi	0.000 psi	6.5E-003 psi	0.03 psi
0.00000 kPa	0.0000 kPa	0.00000 kPa	2.1E+001 Pa	0.2068 kPa
-5.014 psi	-5.02 psi	-0.006 psi	6.5E-003 psi	0.03 psi

Kalibrasi Pressure

1. Lepas Pressure gauge dan pastikan tertutup rapat di needle valve
2. Sambungkan pressure gauge tersebut ke kalibrator
3. Amati dan catat nilai perubahan pressure tersebut



Kalibrasi Pressure

- 4. Pasang kembali Pressure gauge dan pastikan needle valve kembali dibuka**
- 5. Lakukan analisis data dan nilai erornya**

Estimasi Standar	Actual Standar	Pembacaan Alat			Mean	
		Bar	Bar	Bar	Bar	
Bar	Bar	N1	T1	N2		
0.000	0.000	0.00	0.00	0.00	0	
1.000	1.001	1.00	1.00	1.00	1	
3.000	3.002	2.90	3.00	2.90	2.95	
5.000	5.000	4.90	4.90	4.90	4.9	
7.000	7.003	7.00	7.00	6.90	6.975	
		Equipment Reading	Nilai Aktual	Uncertainty	Koreksi	Result
				95%		
		Bar	Bar	Bar	Bar	
		0.000	0.000	0.100	0.000	
		1.000	1.001	0.100	0.001	
		3.000	3.052	0.151	0.052	
		5.000	5.100	0.100	0.100	
		7.000	7.028	0.151	0.028	



Fluke Pressure Oil deadweight tester 700



Prosonic Flow 93T Portable



